



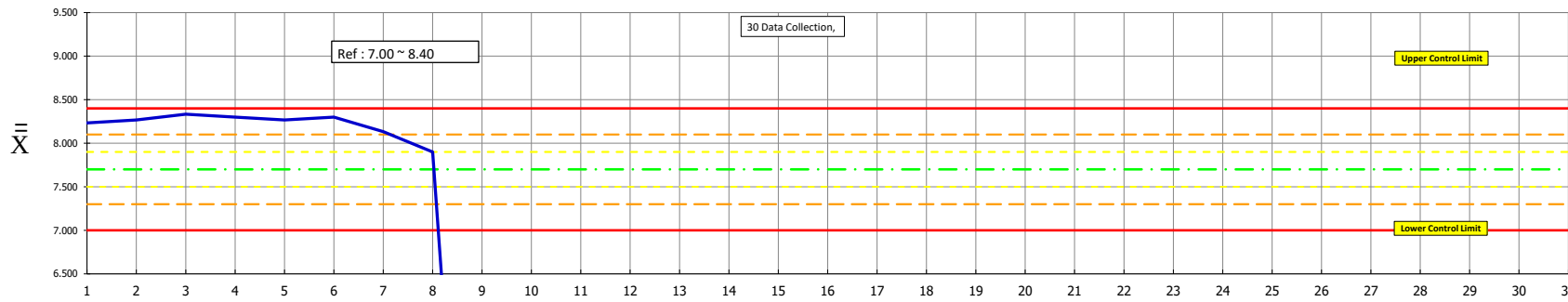
PT.SUMITOMO ELECTRIC WINTEC INDONESIA

No. Dokumen	4-QAS-PP07-001
Revisi	00
Tgl Berlaku	2-Dec-16

 $\bar{X}$ - R CONTROL CHART

MACHINE DF Tester (New)	SECTION QC	OPERATION DF 280° C	DATE CONTROL LIMITS CALC.	SPECIFICATION min 2 max 20	COURSE NO.	CONTROL ITEM (Δ) YES ☐ NO ☐
	MONTH May-25	CHARACTERISTIC Temperatur		SAMPLE SIZE / FREQUENCY n = 3	TYPE / SIZE : D2IDW4-B-2 / 0.70 PIC : DWI AS	

$\bar{X}$ = AVERAGE $\bar{X}$ = **7.700** U.C.L. = $\bar{X} + A_2 R$ = **8.4000** L.C.L. = $\bar{X} - A_2 R$ = **7.0000**
AVERAGE ($\bar{X}$ BAR CHART)

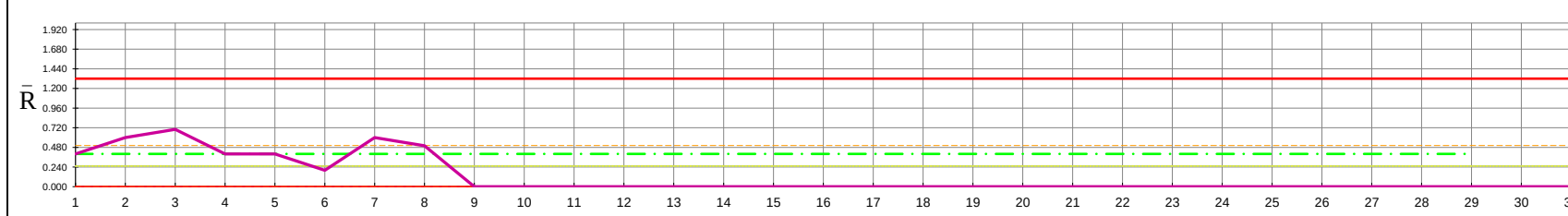


SPECIAL CAUSES

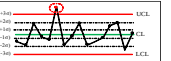
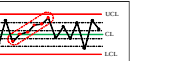
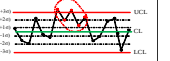
ACTION INSTRUCTION

- _____
- _____
- _____
- _____
- _____

$\bar{R}$ = AVERAGE $\bar{R}$ = **0.400** U.C.L. = $D_4 \bar{R}$ = **1.320** L.C.L. = $D_3 \bar{R}$ = **0.00**
RANGES ($\bar{R}$ - CHART)

Special Cause Marker Legend:
Test 1 • Test 2 ♦ Test 3 ▲ Test 4 ■

SUBGROUP SIZE	A ₂	d ₂	D ₃	D ₄
2	1.881	1.128	0	3.267
3	1.023	1.693	0	2.574
4	0.729	2.059	0	2.281
5	0.577	2.326	0	2.135
6	0.483	2.534	0	2.004
7	0.4193	2.704	0.0756	1.9244
8	0.3726	2.847	0.1361	1.8639

Special Cause	Titik berada di luar garis merah atau lebih dari 3 σ  points more than 3σ from centerline (CL) 6 titik berada dalam satu urutan secara terus menerus, baik posisi naik ataupun turun  6 points in a row, all increasing or all decreasing 4 atau 5 titik lebih dari 1 σ dengan posisi yang sama dan titik keluar dari garis merah atau 3 σ  4 out of 5 points > 1 σ above centerline (same side)																														
	Month																														
Date	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20	21	22	23	24	25	26	27	28	29	30	31
DATA	1	8.400	8.200	8.000	8.100	8.500	8.200	7.800	8.200																						
	2	8.300	8.000	8.700	8.500	8.100	8.300	8.200	7.800																						
	3	8.000	8.600	8.300	8.300	8.200	8.400	8.400	7.700																						
N	3	3	3	3	3	3	3	3	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
AVERAGE, $\bar{X}$	8.233	8.267	8.333	8.300	8.267	8.300	8.133	7.900	#DIV/0!	#DIV/0!	#DIV/0!	#DIV/0!	#DIV/0!	#DIV/0!	#DIV/0!	#DIV/0!	#DIV/0!	#DIV/0!	#DIV/0!	#DIV/0!	#DIV/0!	#DIV/0!	#DIV/0!	#DIV/0!	#DIV/0!	#DIV/0!	#DIV/0!	#DIV/0!	#DIV/0!	#DIV/0!	#DIV/0!
Range	0.400	0.600	0.700	0.400	0.400	0.200	0.600	0.500																							
Moving Range		0.002	0.768	0.000	0.000	0.000	0.000	0.000	0.000	0.000	0.000	0.000	0.000	0.000	0.000	0.000	0.000	0.000	0.000	0.000	0.000	0.000	0.000	0.000	0.000	0.000	0.000	0.000	0.000	0.000	
SHIFT 1		dwas	dwas	dwas			dwas																								
SHIFT 2																															

CALCULATION

$$\begin{aligned} \bar{X} &= 8.2167 \\ R &= 0.4750 \\ S_{\bar{X}} &= 0.2806 \quad S_{\bar{X}} = 0.251 \\ \text{THE PROCESS MUST BE IN CONTROL BEFORE CAPABILITY CAN BE DETERMINED} \\ C_{pk} &= \min \left[\frac{\bar{X} - LSL}{3S_{\bar{X}}}, \frac{USL - \bar{X}}{3S_{\bar{X}}} \right] \\ &= \min [1.45 \quad 0.22] \\ &= 0.22 \\ Cp &= \frac{USL - LSL}{6S_{\bar{X}}} \\ &= [0.83] \end{aligned}$$

CHECKED BY _____ APPROVED BY _____